

The Predictive Power of Eye-Tracking Data in an Interactive AR Learning Environment

David Dzsotjan
dzsotjan@rhrk.uni-kl.de
Department of Physics, TU
Kaiserslautern
Kaiserslautern, Germany

Kim Ludwig-Petsch
k.ludwig-petsch@deutsches-
museum.de
Deutsches Museum
Munich, Germany

Sergey Mukhametov
mukhamet@physik.uni-kl.de
Department of Physics, TU
Kaiserslautern
Kaiserslautern, Germany

Shoya Ishimaru
Shoya.Ishimaru@dfki.de
TU Kaiserslautern & DFKI GmbH
Kaiserslautern, Germany

Stefan Küchemann
s.kuechemann@physik.uni-kl.de
Department of Physics, TU
Kaiserslautern
Kaiserslautern, Germany

Jochen Kuhn
kuhn@physik.uni-kl.de
Department of Physics, TU
Kaiserslautern
Kaiserslautern, Germany

ABSTRACT

Learning through embodiment is a promising concept, potentially capable to remove many layers of abstraction hindering the learning process. Walk the Graph, our HoloLens2-based AR application, provides an inquiry-based learning setting for understanding graphs through the full-body movement of the user. In this paper, as part of our ongoing work to build an AI framework to quantify and predict the learning gain of the user, we examine the predictive potential of gaze data collected during the app usage. To classify users into groups with different learning gains, we construct a map of areas of interest (AOI) based on the gaze data itself. Subsequently, using a sliding window approach, we extract engineered features from the collected in-app as well as gaze data. Our experimental results have shown that a Support Vector Machine with selected features achieved the highest F1 score (0.658; baseline: 0.251) compared to other approaches including a K-Nearest Neighbor and a Random Forest Classifier although in each of the cases the lion's share of the predictive power is indeed provided by the gaze-based features.

CCS CONCEPTS

• **Applied computing** → **Interactive learning environments.**

KEYWORDS

Inquiry-based Learning; Augmented Reality; Eye Tracking.

ACM Reference Format:

David Dzsotjan, Kim Ludwig-Petsch, Sergey Mukhametov, Shoya Ishimaru, Stefan Küchemann, and Jochen Kuhn. 2021. The Predictive Power of Eye-Tracking Data in an Interactive AR Learning Environment. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-8461-2/21/09...\$15.00

<https://doi.org/10.1145/3460418.3479358>

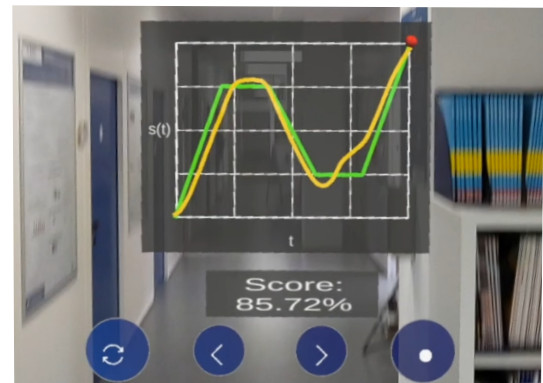


Figure 1: First-person view of the UI for Walk the Graph. The original (green) and the user-produced (yellow) graphs are traced on the virtual HUD and the red dot indicates the position of the user. The virtual buttons allow for interaction with the app through hand tracking and gesture recognition.

September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages.
<https://doi.org/10.1145/3460418.3479358>

1 INTRODUCTION

1.1 The significance of graphs

The concept and use of graphs and diagrams are fundamental for learning and problem solving in all STEM disciplines [1]. Graphs are especially important for simplifying complex concepts [4], visualizing measurement data [3], for the exchange of information between individuals [13] and for online reasoning [15]. A convention strictly determines the way the data is visualized in a graph, and therefore, there is a clear procedure how to extract information from it [5, 12]. However, a deep understanding of graphs often poses a challenge to students, especially concerning the more abstract properties, like the slope or the area under the curve [14].

1.2 Eye tracking for graph interpretation

Eye tracking is an excellent tool to quantify the information extraction and processing strategies of learners and has been widely

used in education research to investigate students' interactions with graphs [2, 7–9, 11, 14]. In this series of works, Susac et al. studied the visual attention during problem-solving of items in which physics and non-physics students had to determine the slope of a graph or the area underneath a graph in different contexts [14]. The authors found that physics students achieved higher scores in these graph tasks and focused significantly longer on the graph area than non-physics students. In a replication study of this work, Klein et al. found no difference in the total attention time concentrated on the graph area between physics and non-physics students [8]. Instead, students who determined the slope of a graph correctly focused their gaze significantly longer along the graph.

This observation was also corroborated by Küchemann et al. who analyzed the widely-used Test of Understanding Graphs in Kinematics (TUG-K) [11]. The authors found that students who determine the slope of a graph in a physics context correctly focused significantly longer along the graph, and students who determine the area underneath the graph correctly also focused longer along the graph and underneath the graph.

Despite a vast amount of eye-tracking research centered on graphs in a problem-solving context, as of now, the number of articles targeting visual attention on graphs during learning is tiny in comparison. In a recent systematic literature review about students' visual processes during problem-solving and learning with graphs until 2020 Küchemann et al. could only identify two articles that focus on the visual attention on graphs during learning [10]. The authors found that the processing of graphs during learning in comparison to a problem-solving scenario is significantly different and the results from problem solving cannot be transferred to learning processes per se. However, the monitoring of students' graph understanding during the learning process is essential to provide tailored real-time support and to enhance the learning outcome.

The current work contributes to filling this research gap by investigating the role of eye movements during an embodied learning experience in an augmented-reality learning environment.

1.3 Research questions

In order to diminish the degree of abstraction and make the concept of graphs more tangible upon learning, we have developed *Walk the Graph*, an AR application which provides users with a compact, inquiry-based learning environment for understanding graphs through full-body movement (see Figure 1). By moving in real space, the learners have to reproduce kinematic graphs projected into their field of view on a virtual head-up display (HUD). A Microsoft HoloLens2 is used as a wearable AR platform for the application, offering the advantages of mixed reality and an eye-tracking functionality.

The identification of indicators that are predictive for the successful learning process in real-time is an essential aspect to provide an adaptive learning support. We are currently developing an AI framework to reliably assess and predict the learning behavior of the app user, based on in-app performance as well as on the collected gaze data, the latter being a major source of information for prediction. Logically, it is vital to establish whether the collected eye-tracking data contains sufficient information to do that. We conducted an experiment with 36 participants to answer our research

question: *Is students' gaze behavior during learning with Walk the Graph predictive in regard to their learning gain?*

In summary, contributions in this paper include: (1) a demonstration of HoloLens2-based AR application which helps students to learn the concept and use of graphs through full-body movements and (2) an experimental result showing the potential of eye gaze while using the app for predicting students' learning gain.

2 APP DESIGN

The aim of the application *Walk the Graph* is to teach the concept of graphs and diagrams through embodiment: the user physically experiences the mapping between real-world movement and kinematic graphs. We have developed this app for the Microsoft HoloLens2, using the programming language C#, with Unity as the mixed-reality engine. The application presents the user with different position-time kinematic graphs which they are supposed to reproduce with their own physical motion as accurately as possible, by walking along a predefined line in the room. The position of the user is captured by the inertial measurement unit within the HoloLens2 as well as its spatial mapping functionality.

As shown in Figure 1, the user interface includes a virtual screen that contains the original graph and virtual buttons for navigation and starting the measurement, the user interacts with these through hand tracking and gesture recognition. The app tracks and shows the current position of the user, as well as the graph that their motion has mapped out, superimposed on the original graph. After every attempt, the user performance is quantified in percentage, showing how close the "walked" graph to the original was.

In order to harness the eye-tracking capability of the HoloLens2 and be able to record user gaze behavior during the app usage, we have embedded the plugin developed by Kapp et al. [6]. This, by using a web server, makes it possible to operate the gaze data collection from another device, thereby eliminating a potential distraction for the user. It is especially useful in case of a study since the probability of mistakes in the data recording is greatly reduced by this design.

3 METHODS

3.1 Target groups, populations and walk phases

The results presented in this paper are based on data gathered during our pilot study in the Deutsches Museum Munich, in November 2020. Here, after having completed a pre-test containing graph-related problems, the participants used the *Walk the Graph* application, and subsequently filled out a post-test that assessed whether there has been any change in the understanding of graph-related concepts compared to the pre-test. We have divided the participants of the pilot study into three groups: Positive learning gain (those with better scores in the post-test compared to the pre-test, 10 people), Non-positive learning gain (those who did not show any improvement or their performance even decreased, 12 people), and Experts (maximum points achieved in both pre- and post-tests, 14 people). In the following, we perform a binary classification, attempting to predict whether a user belongs in the positive learning gain group or not, thereby demonstrating the importance of the eye-tracking data.

Using the app, each run is separated into 3 distinct phases: “planning”, where the user has not started the measurement yet but can already see the graph on the virtual HUD, “walk”, where the users full-body movement is recorded and traced on the user interface and “assessment”, where the measurement is over but the user can still see the original graph and their own result. Here, we will examine and use the data from the “planning” phase.

3.2 Composite AOI map and features for prediction

A handy way to translate the time series of recorded gaze points into predictors is to define areas of interest (AOI) in the image and engineer features based on the gaze dwell times in the respective AOIs. Deviating from the usual way where AOIs are set around key objects on the graph (origin, axes, points on the graph where the slope changes, etc), we let the data itself dictate what areas have a higher importance.

Figure 2 is an example for such a map of AOIs. It is obtained by creating gaze heatmaps for the separate learning groups (taking the individual gaze points, fitting Gaussian functions on them and normalizing the manifold). Subsequently, we take a linear combination of these heatmaps and apply a threshold, under which we set the heatmap values to zero. As a result, the emerging “islands” represent areas to which members of the different groups typically direct their gaze. Thus, this composite AOI map is an excellent basis for classifying individuals into different learning groups.

We subsequently engineer the manifold of features as total and mean attention times, as well as their standard deviations, related to each AOI, scaled by the total time for the planning phase. We apply a sliding window method (with window size and step size of 0.3 and 0.1, respectively, in units of the planning phase duration) for obtaining the data samples. This increases the size of the datasets as well as it introduces a time evolution dimension. As non-eye-tracking features, we include the in-app scores for the individual graphs and attempts, the attempt number, the duration of the planning phase, as well as the time evolution parameter, related to the sliding window position.

4 RESULTS

4.1 Feature selection

It is often said that too much information is no information - and this rings true in our case as well. It is important to check for the presence of correlations between features or simply noisy, non-predictive features that potentially lead to the decrease of prediction efficiency. To order our 104 features according to relevance, we examined how well an estimator (a Support Vector Machine algorithm with the Radial Basis Function kernel) performs a binary classification between the positive learning-gain group and the rest of the participants, in case of different subsets of the existing features. Starting out with a single feature, and, in every iteration, adding the feature that leads to the highest F1 score upon prediction, we established an order of features as per relevance. In each iteration, the F1 score was calculated by tuning the hyperparameters of the classifier, so that the average of the F1 scores over a 3-fold cross-validation would be maximal.

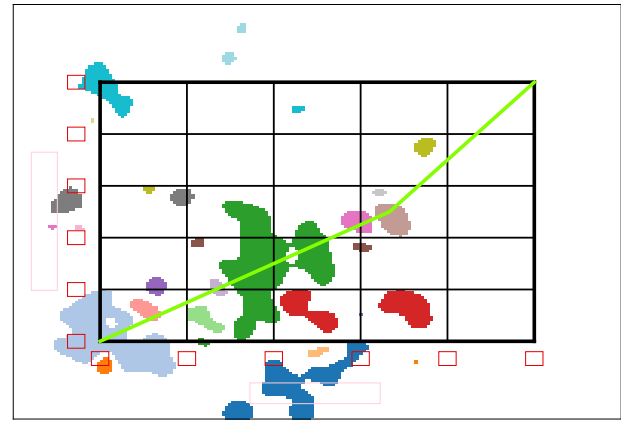


Figure 2: Example for a data-driven composite AOI map: taking the linear combination of gaze heatmaps of the 3 separate learning gain groups and applying a high-pass filter to it yields a manifold of separate areas at which members of the different learning groups typically look.

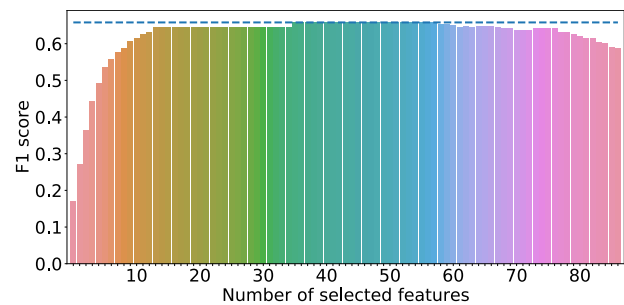


Figure 3: Cross-validated F1 score as we add the next best feature to the list of features used for training a support vector machine classifier with an RBF kernel. This procedure sorts features according to their predictive power.

Figure 3 shows the cross-validated F1 score at every iteration, as the next best feature was chosen. From 1 to about 35 it grows and reaches its maximum, however, adding more features leads to the deterioration of the F1 score. Thus, for model training, we use the restricted feature manifold that provides the maximal F1 score.

4.2 Predictive power of the gaze data

By now we have everything we need to gauge the predictiveness of the eye-tracking data. For the sake of comparison, we have taken three sets of features: the non-eye-tracking features (i.e., those not derived from the gaze data, like in-app score, attempt number, total phase duration, etc.); purely eye-tracking features (a set of features that are exclusively eye-tracking based and are also taken from the restricted feature set mentioned in the previous subsection, these include mainly total and mean dwell times in AOIs as well as standard deviation thereof); and lastly, the restricted feature manifold consisting of a mixed set of features.

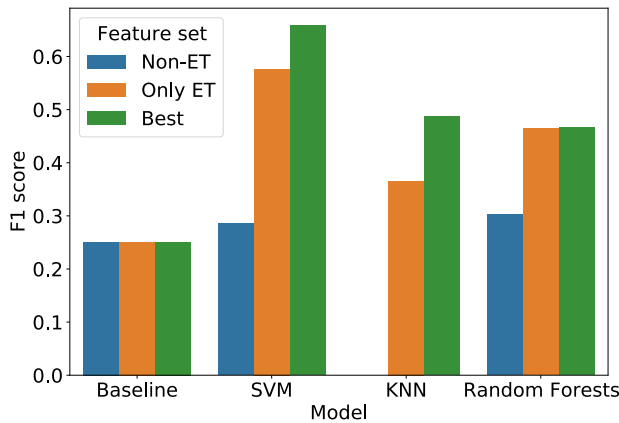


Figure 4: Cross-validated F1 scores in case of a set of non-gaze type of features, only gaze-data-based features and the restricted set of features, for support vector machines, K-nearest neighbors and random forests classifiers, compared with a probabilistic baseline estimator.

It is important to note that throughout our investigation, since in the dataset several samples belong to the same user, upon constructing our cross-validation folds, we have kept train and validation sets independent by not letting data from the same user get into both sets within the same fold. Also, we used a stratified sampling of true/false labelled samples, that is, the true/false ratio in the train and validation sets is kept as similar as possible.

As depicted in Figure 4, we have trained several classifiers on these three feature manifolds and compared the resulting cross-validated F1 scores. With an adaptive grid search, we have tuned the hyperparameters of a support vector machines classifier, a K-nearest neighbors classifier and a random forests classifier, and have compared them with a baseline classifier that predicts “true” with a probability that equals the true/false ratio in the training set. The numerical values for the respective F1 scores are presented in Table 1.

5 DISCUSSION

We have seen that in order to bring out the maximal predictive potential from our dataset, it is important to examine the features and keep only those that contribute positively to the model quality. Looking at Figure 3 more attentively, one also sees signs of non-trivial mechanisms. For one, we observe clear signs of not only a saturation in the F1 score but a decrease as we add more features. The potential reasons for this are correlations between features (for example, mean and total attention durations can very possibly be correlated), as well as some features simply not being predictive. The correlations could be effectively eliminated by reducing the dimensions of the data using principal component analysis.

The other intriguing feature is the presence of plateaus: instead of a monotonous growth in F1 until its maximum. This points to the interesting phenomenon that some features by themselves do not increase the prediction score, but a certain combination of them together contributes to the stepwise increase.

Table 1: Cross-validated F1 scores for the classifiers (baseline, support vector machines, K-nearest neighbors, random forests), each trained with non-eye-tracking, purely eye tracking and the restricted (best) set of features.

Model	Non-ET feat.	Only ET feat.	Best feat.
Baseline	0.251	0.251	0.251
SVM	0.285	0.576	0.658
KNN	0.000	0.365	0.487
Random Forests	0.302	0.464	0.466

The results presented in Figure 4 answer our research question, i.e., whether gaze-data-based features have the potential to be good predictors for the learning behavior. As we can see, using the non-eye-tracking features, all classifiers are just marginally better than the baseline estimator - except the K-nearest neighbors estimator, where the prediction score is zero. The use of purely eye-tracking features results in a sharp improvement in the performance for each model. Taking the restricted set of features gives the best scores, but the increase is not as dramatic as the score difference between non-eye-tracking and exclusively eye-tracking-based features. According to our results, we can safely say that gaze-based data is a good basis for predicting learning behavior in users of Walk the Graph.

6 CONCLUSION AND OUTLOOK

In this paper, we have presented our investigation about the usefulness of eye-tracking data recorded during learning for predicting the learning gain of the user. The data used here was generated at our pilot study by participants using our Walk the Graph app designed for the Microsoft HoloLens2 where the users had to reproduce kinematic graphs with their own movement. Based on it, we have performed binary classification to predict whether individual instances correspond to a positive learning gain or not. Having engineered features based on a data-generated AOI map, we could isolate the features that maximize the cross-validated F1 prediction score. Subsequently, we have trained several classifiers on different set of features (including and excluding eye-tracking based features, respectively), and found that gaze-based features have a substantial contribution to predicting the learning gain.

The presented approach could have great potential for constructing adaptive learning systems in the fields of STEM education, such as adaptive digital textbooks or interactive learning tools. Use cases outside of education are also very conceivable: our findings could be used in medicine, for example, in movement therapy and for recovery of functional motor capabilities; or in the context of sport, especially as a support for training measures in athletics.

This result, apart from its broader scientific significance, is also an important step for the subsequent development of our project, since we aim to use eye-tracking data as a major resource for assessing and predicting learning behavior in users. Future steps include the construction of a machine learning framework that makes use of all phases of the run, that is: planning, walk and assessment, and provides a single prediction for the learning gain of the user. We also plan to conduct a full study in order to assess the impact of

the degree of embodiment (that is, the freedom of motion) on the learning gain. Our ultimate goal is to construct a system that can predict the learning gain of the user on the fly and offer adaptive assistance if needed (with graphical or text-based hints), thereby facilitating the learning process in a personalized manner.

REFERENCES

- [1] G. Michael Bowen and Wolff-Michael Roth. 1998. Lecturing graphing: What features of lectures contribute to student difficulties in learning to interpret graph? *Research in Science Education* 28, 1 (01 Mar 1998), 77. <https://doi.org/10.1007/BF02461643>
- [2] Sebastian Brückner, Olga Zlatkin-Troitschanskaia, Stefan Küchemann, Pascal Klein, and Jochen Kuhn. 2020. Changes in Students' Understanding of and Visual Attention on Digitally Represented Graphs Across Two Domains in Higher Education: A Postreplication Study. *Frontiers in Psychology* 11 (2020), 2090. <https://doi.org/10.3389/fpsyg.2020.02090>
- [3] Bronwen Cowie and Beverley Cooper. 2017. Exploring the challenge of developing student teacher data literacy. *Assessment in Education: Principles, Policy & Practice* 24, 2 (2017), 147–163. <https://doi.org/10.1080/0969594X.2016.1225668> arXiv:<https://doi.org/10.1080/0969594X.2016.1225668>
- [4] Frances R. Curcio. 1987. Comprehension of Mathematical Relationships Expressed in Graphs. *Journal for Research in Mathematics Education JRME* 18, 5 (1987), 382–393. <https://doi.org/10.5951/jresmetheduc.18.5.0382>
- [5] Joseph Goldberg and Jonathan Helfman. 2011. Eye tracking for visualization evaluation: Reading values on linear versus radial graphs. *Information Visualization* 10, 3 (2011), 182–195. <https://doi.org/10.1177/1473871611406623> arXiv:<https://doi.org/10.1177/1473871611406623>
- [6] Sebastian Kapp, Michael Barz, Sergey Mukhametov, Daniel Sonntag, and Jochen Kuhn. 2021. ARETT: Augmented Reality Eye Tracking Toolkit for Head Mounted Displays. *Sensors* 21, 6 (2021). <https://doi.org/10.3390/s21062234>
- [7] Pascal Klein, Sebastian Becker, Stefan Küchemann, and Jochen Kuhn. 2021. Test of understanding graphs in kinematics: Item objectives confirmed by clustering eye movement transitions. *Phys. Rev. Phys. Educ. Res.* 17 (Mar 2021), 013102. Issue 1. <https://doi.org/10.1103/PhysRevPhysEducRes.17.013102>
- [8] Pascal Klein, Stefan Küchemann, Sebastian Brückner, Olga Zlatkin-Troitschanskaia, and Jochen Kuhn. 2019. Student understanding of graph slope and area under a curve: A replication study comparing first-year physics and economics students. *Phys. Rev. Phys. Educ. Res.* 15 (Aug 2019), 020116. Issue 2. <https://doi.org/10.1103/PhysRevPhysEducRes.15.020116>
- [9] Maria Kozhevnikov, Michael A. Motes, and Mary Hegarty. 2007. Spatial Visualization in Physics Problem Solving. *Cognitive Science* 31, 4 (2007), 549–579. <https://doi.org/10.1080/15326900701399897>
- [10] Stefan Küchemann, Nils Cullmann, Sarah Kovac, Sebastian Becker, Pascal Klein, Kathrin Kennel, Stefan Ruzika, and Jochen Kuhn. 2021. *Blickverhalten beim Lernen und Problemlösen mit Graphen: ein Literaturüberblick bis 2020*. Springer Spektrum, Berlin, Heidelberg. accepted.
- [11] Stefan Küchemann, Pascal Klein, Sebastian Becker, Niharika Kumari, and Jochen Kuhn. 2020. Classification of Students' Conceptual Understanding in STEM Education using Their Visual Attention Distributions: A Comparison of Three Machine-Learning Approaches.. In *CSEdu* (1). 36–46.
- [12] David Peebles and Peter C-H Cheng. 2003. Modeling the effect of task and graphical representation on response latency in a graph reading task. *Human factors* 45, 1 (2003), 28–46.
- [13] Benjamin Strobel, Marlit Annalena Lindner, Steffani Saß, and Olaf Köller. 2018. Task-irrelevant data impair processing of graph reading tasks: An eye tracking study. *Learning and Instruction* 55 (2018), 139–147.
- [14] Ana Susac, Andreja Bubic, Elizabeta Kazotti, Maja Planinic, and Marijan Palmovic. 2018. Student understanding of graph slope and area under a graph: A comparison of physics and nonphysics students. *Physical Review Physics Education Research* 14, 2 (2018), 020109.
- [15] Sam Wineburg, Joel Breakstone, Sarah McGrew, and Teresa Ortega. 2018. *Why Google Can't Save Us*. Springer Fachmedien Wiesbaden, Wiesbaden, 221–228. https://doi.org/10.1007/978-3-658-19567-0_13